

Appendix A: Full API Documentation

This appendix documents every endpoint exposed by the Accident Data API. For each endpoint it lists the HTTP method and path, all parameters with their types and meaning, the structure of the returned object, the relevant HTTP status codes, and an example request and response. The same information is also available interactively through the Swagger UI at /docs once the FastAPI service is running.

A.1 Metadata Endpoints

1. GET /

Returns basic information about the API, including its current status and a link to the documentation. Used as a health check and entry point.

Parameters

None.

Response fields (RootInfo)

Field	Type	Description
name	string	Service name ("DBW Accident Data API").
version	string	API version string, e.g. "1.0".
status	string	Service status, e.g. "running".
documentation	string	Relative path to the interactive documentation ("/docs").

Status codes

Status	Meaning
200	Service is reachable; information returned successfully.

Example response

```
{ "name": "DBW Accident Data API", "version": "1.0", "status": "running",  
  "documentation": "/docs" }
```

2. GET /metadata/sources

Returns metadata about every external dataset integrated into the system, including its type, original URL, and licence. Used to expose data provenance to API consumers.

Parameters

None.

Response fields (list of SourceItem)

Field	Type	Description
source_id	integer	Internal identifier for the data source.
name	string	Human-readable name of the dataset, e.g. "Unfallatlas".
source_type	string	Type of source, e.g. CSV download or REST API.

url	string	Original URL the dataset was retrieved from.
license	string	Licence under which the dataset is published.

Status codes

Status	Meaning
200	Source metadata returned successfully.
500	Database error while retrieving source information.

Example response

```
[ { "source_id": 1, "name": "Unfallatlas", "source_type": "CSV", "url": "https://www.opengeodata.nrw.de/...", "license": "dl-by-de/2.0" } ]
```

3. GET /import-runs

Returns the history of ETL import operations, including which source was loaded, when, how many records were imported, and the resulting status. Used to demonstrate reproducibility and auditability of the data pipeline.

Parameters

None.

Response fields (list of ImportRunItem)

Field	Type	Description
run_id	integer	Identifier of the import run.
source_name	string	Name of the dataset that was imported.
started_at	datetime	Timestamp the import run started.
finished_at	datetime	Timestamp the import run finished.
records_imported	integer	Number of rows written during this run.
status	string	Outcome of the run, e.g. "success" or "failed".
notes	string	Free-text notes about the run.

Status codes

Status	Meaning
200	Import history returned successfully.
500	Database error while retrieving import information.

Example response

```
[ { "run_id": 4, "source_name": "Unfallatlas", "started_at": "2026-01-10T08:00:00", "finished_at": "2026-01-10T08:04:12", "records_imported": 91423, "status": "success", "notes": null } ]
```

A.2 Counts and Filters Endpoints

1. GET /accidents/filter-by-region

Returns the number of accidents for one region at a chosen level (state, district, or municipality) and year.

Parameters

Parameter	Type	Required	Description
level	enum: state district municipality	Yes	Region granularity to filter by.
code	string	Yes	AGS code identifying the region, e.g. "01" (Schleswig-Holstein) or "01002" (Kiel).
year	integer	No (default 2023)	Accident year to filter by.

Response fields (RegionCountResponse)

Field	Type	Description
metadata	object	Licence metadata (name, code, URL) attached to every response.
level	string	Echo of the requested region level.
code	string	Echo of the requested AGS code.
year	integer	Echo of the requested year.
accident_count	integer	Number of accidents matching the filter.

Status codes

Status	Meaning
200	Count returned successfully (zero is a valid count).
500	Database error while counting accidents.

Example request

```
GET /statistics/accidents-by-region?level=district&code=01002&year=2023
```

Example response

```
{ "metadata": {...}, "level": "district", "code": "01002", "year": 2023, "accident_count": 1290 }
```

2. **GET** /accidents/filter-by-time

Returns the number of accidents matching any combination of year, month, weekday, and hour.

Parameters

Parameter	Type	Required	Description
year	integer	No	e.g. 2023.
month	integer (1–12)	No	Month of the year.
weekday	integer	No	Weekday code as stored in the source data.
hour	integer (0–23)	No	Hour of the day.

Response fields (TimeFilterResponse)

Field	Type	Description
metadata	object	Licence metadata.
filters	object	Echo of the applied year / month / weekday / hour filters.
accident_count	integer	Number of accidents matching all supplied filters.

Status codes

Status	Meaning
200	Count returned successfully.
500	Database error while processing the request.

Example request

GET /accidents/filter-by-time?year=2023&month=12

3. GET /accidents/filter-by-category

Returns the number of accidents matching year, federal state, severity category, and/or participant type. This is the endpoint used to answer most of the mandatory examiner questions (e.g. accidents in Saxony in 2023, pedestrian accidents in Berlin in 2023).

Parameters

Parameter	Type	Required	Description
year	integer	No	e.g. 2023.
state_code	string	No	Two-digit AGS state prefix (e.g. "14" = Saxony), or a longer district/municipality prefix.
accident_category	enum: 1 2 3	No	1 = Fatal, 2 = Serious injury, 3 = Minor injury.
participant	enum: pedestrian bicycle motorcycle car	No	Restricts the count to accidents involving this participant type.

Response fields (FilterByCategoryResponse)

Field	Type	Description
metadata	object	Licence metadata.
filters	object	Echo of year, state_code, state_name, accident_category, participant.
accident_count	integer	Number of accidents matching all supplied filters.

Status codes

Status	Meaning
200	Count returned successfully.

400	Invalid filter values supplied.
500	Database error while processing the request.

Example request

GET /accidents/filter-by-category?state_code=11&year=2023&participant=pedestrian

Example response

```
{ "metadata": {...}, "filters": { "year": 2023, "state_code": "11", "state_name": "Berlin", "accident_category": null, "participant": "pedestrian" }, "accident_count": 742 }
```

A.3 Time Analytics Endpoints

1. GET /statistics/earliest-year

Returns the earliest accident year available anywhere in the dataset, independent of region.

Parameters

None.

Response fields (EarliestYearResponse)

Field	Type	Description
metadata	object	Licence metadata.
earliest_year	integer null	First year for which any accident record exists.

Status codes

Status	Meaning
200	Earliest year returned successfully.
500	Database error while retrieving the earliest year.

2. GET /statistics/state-earliest/{state_code}

Returns the earliest accident year available for one specific federal state. Used to answer questions such as “from which year is data available for North Rhine-Westphalia?”

Parameters

Parameter	Type	Required	Description
state_code	string (path)	Yes	Two-digit AGS state code, e.g. “05” = North Rhine-Westphalia, “13” = Mecklenburg-Western Pomerania.

Response fields (StateEarliestResponse)

Field	Type	Description
metadata	object	Licence metadata.
state_code	string	Echo of the requested state code.
state_name	string	Resolved state name.

earliest_year	integer	First year with accident data for this state.
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Status codes

Status	Meaning
200	Earliest year for the state returned successfully.
404	Unknown state code, or no accident data exists for that state.
500	Database error while retrieving the data.

Example request

```
GET /statistics/state-earliest/05
```

Example response

```
{ "metadata": {...}, "state_code": "05", "state_name": "Nordrhein-Westfalen",
"earliest_year": 2016 }
```

3. **GET** /statistics/year-over-year/{state_code}

Returns the accident count for one federal state for every available year, together with the year-on-year change. Used for trend analysis.

Parameters

Parameter	Type	Required	Description
state_code	string (path)	Yes	Two-digit AGS state code (01–16).

Response fields (YearOverYearResponse)

Field	Type	Description
metadata	object	Licence metadata.
state_code	string	Echo of the requested state code.
state_name	string	Resolved state name.
trend_summary	list of TrendPoint	One entry per year: { year, accident_count, change_from_previous_year }.

Status codes

Status	Meaning
200	Trend data returned successfully.
404	Unknown state code, or no data exists for that state.
500	Database error while building the trend.

A.4 Rates and Rankings Endpoints

1. **GET** /statistics/rates/{year}

Returns the regions with the highest accidents-per-10,000-inhabitants rate for a given year, based on the official indicator dataset.

Parameters

Parameter	Type	Required	Description
year	integer (path)	Yes	Year to retrieve rates for.
limit	integer	No (default 10)	Maximum number of regions to return.

Response fields (AccidentRatesResponse)

Field	Type	Description
metadata	object	Licence metadata.
year	integer	Echo of the requested year.
results	list of RegionRate	Each entry: { ags, region_name, year, accidents_per_10000_inhabitants }.

Status codes

Status	Meaning
200	Rates returned successfully.
404	No rate data available for the requested year.
500	Database error while retrieving rates.

Example request

GET /statistics/rates/2023?limit=5

Example response

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/statistics/rates/2023?limit=1' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/statistics/rates/2023?limit=1
```

Server response

Code	Details
200	Response body <pre>{ "metadata": { "license_name": "Datenlizenz Deutschland - Namensnennung - Version 2.0 (DL-DE BY 2.0)", "license_code": "dl-by-de/2.0", "license_url": "http://www.dcat-ap.de/def/licenses/dl-by-de/2.0" }, "year": 2023, "results": [{ "ags": "07318", "region_name": "Speyer, kreisfreie Stadt", "year": 2023, "accidents_per_10000_inhabitants": 79.4 }] }</pre>

Response headers

```
content-length: 330
content-type: application/json
date: Thu, 25 Jun 2026 18:39:46 GMT
server: uvicorn
```

2. **GET** /statistics/district-rankings/{year}

Returns the top or bottom districts ranked by total accident count for a given year.

Parameters

Parameter	Type	Required	Description
year	integer (path)	Yes	Year to rank districts for.
order	enum: top bottom	No (default top)	Ranking direction.
limit	integer, 1–50	No (default 10)	Number of districts to return.

Response fields (DistrictRankingsResponse)

Field	Type	Description
metadata	object	Licence metadata.
year	integer	Echo of the requested year.
ranking_type	string	Echo of “top” or “bottom”.
results	list of DistrictRankItem	Each entry: { district_code, district_name, year, accident_count }.

Status codes

Status	Meaning
200	Ranking returned successfully.
404	No district data available for the requested year.

Example of request & response

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/statistics/district-rankings/2023?order=top&limit=2' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/statistics/district-rankings/2023?order=top&limit=2
```

Server response

Code	Details
200	Response body<pre>{ "metadata": { "license_name": "Datenlizenz Deutschland – Namensnennung – Version 2.0 (DL-DE BY 2.0)", "license_code": "dl-by-de/2.0", "license_url": "http://www.dcat-ap.de/def/licenses/dl-by-de/2.0" }, "year": 2023, "ranking_type": "top", "results": [{ "district_code": "04011", "district_name": "Bremen, Stadt", "year": 2023, "accident_count": 2199 }, { "district_code": "01056", "district_name": "Pinneberg", "year": 2023, "accident_count": 1216 }] }</pre>  Download

Response headers

```
content-length: 423
content-type: application/json
date: Thu, 25 Jun 2026 18:41:51 GMT
server: uvicorn
```


3. GET /statistics/compare-regions

Returns accident counts for several regions across a range of years, for side-by-side comparison.

Parameters

Parameter	Type	Required	Description
region_codes	string	Yes	Comma-separated AGS prefixes, e.g. "14,11,05".
start_year	integer	Yes	First year (inclusive).
end_year	integer	Yes	Last year (inclusive).

Response fields (CompareRegionsResponse)

Field	Type	Description
metadata	object	Licence metadata.
start_year / end_year	integer	Echo of the requested range.
regions	list of RegionComparison	Each entry: { region_code, region_name, values: [{ year, accident_count }] }.

Status codes

Status	Meaning
200	Comparison returned successfully (empty lists are valid).
500	Database error while comparing regions.

Example request

GET /statistics/compare-regions?region_codes=14,11,05&start_year=2018&end_year=2023

Example response

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/statistics/compare-regions?region_codes=14%2C11%2C05&start_year=2018&end_year=2023' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/statistics/compare-regions?region_codes=14%2C11%2C05&start_year=2018&end_year=2023
```

Server response

Code	Details
200	Response body <pre>{ "metadata": { "license_name": "Datenlizenz Deutschland - Namensnennung - Version 2.0 (DL-DE BY 2.0)", "license_code": "dl-by-de/2.0", "license_url": "http://www.dcat-op.de/def/licenses/dl-by-de/2.0" }, "start_year": 2018, "end_year": 2023, "regions": [{ "region_code": "14", "region_name": "Sachsen", "values": [{ "year": 2023, "accident_count": 12513 }] }, { "region_code": "11", "region_name": "Berlin", "values": [{ "year": 2023 }] }] }</pre> Response headers <pre>content-length: 534 content-type: application/json date: Thu, 25 Jun 2026 18:44:20 GMT server: uvicorn</pre>

A.5 Cross-Source Analytics Endpoints

These two endpoints answer questions that cannot be derived from a single source dataset: each combines event-level accident data with the official population-normalised indicator dataset and the AGS region reference.

1. GET /statistics/integrated/fatal-districts/{year}

Returns the districts with the highest number of fatal accidents for a given year, together with their official accidents-per-10,000-inhabitants rate.

Parameters

Parameter	Type	Required	Description
year	integer (path)	Yes	Year to analyse.
limit	integer, 1–50	No (default 5)	Number of districts to return.

Response fields (FatalDistrictsResponse)

Field	Type	Description
metadata	object	Licence metadata.
question	string	Plain-language statement of the question this endpoint answers.
year	integer	Echo of the requested year.
datasets_combined	list of string	Names of the datasets joined to produce this result.

results	list of FatalDistrictResult	Each entry: { rank, district_code, district_name, total_accidents, fatal_accidents, accidents_per_10000_inhabitants }.
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Status codes

Status	Meaning
200	Ranking returned successfully.
404	No matching integrated data for the requested year.
500	Database error during multi-source aggregation.

Example request

GET /statistics/integrated/fatal-districts/2023?limit=2

Example response

The screenshot shows a REST client interface with the following details:

- Curl:** `curl -X 'GET' \ 'http://127.0.0.1:8000/statistics/integrated/fatal-districts/2023?limit=2' \ -H 'accept: application/json'`
- Request URL:** `http://127.0.0.1:8000/statistics/integrated/fatal-districts/2023?limit=2`
- Server response:**
 - Code:** 200
 - Details:** Response body

The response body is a JSON object:

```
{
  "metadata": {
    "license_name": "Datenlizenz Deutschland - Namensnennung - Version 2.0 (DL-DE BY 2.0)",
    "license_code": "dl-by-de/2.0",
    "license_url": "http://www.dcat-op.de/def/licenses/dl-by-de/2.0"
  },
  "question": "Which districts recorded the highest number of fatal accidents, and what was their official accident rate?",
  "year": 2023,
  "datasets_combined": [
    "Unfallatlas accident event data",
    "Regional accident-rate indicator data",
    "AGS/QV-Isys regional reference"
  ],
  "results": [
    {
      "rank": 1,
      "district_code": "01055",
      "district_name": "Ostholstein",
      "total_accidents": 991,
      "fatal_accidents": 13,
      "accidents_per_10000_inhabitants": 57.4
    },
    {
      "rank": 2,
      "district_code": "15085"
    }
  ]
}
```

2. GET /statistics/integrated/bicycle-risk-districts/{year}

Returns the districts with the highest bicycle-accident share for a given year, combined with the official accident-rate indicator. A minimum total-accident threshold prevents small samples from dominating the ranking.

Parameters

Parameter	Type	Required	Description
year	integer (path)	Yes	Year to analyse.
limit	integer, 1–50	No (default 10)	Number of districts to return.
minimum_accidents	integer	No (default 100)	Minimum total accidents a district must have to be included.

Response fields (BicycleRiskResponse)

Field	Type	Description
metadata	object	Licence metadata.
question	string	Plain-language statement of the question this endpoint answers.
year / minimum_accidents	integer	Echo of the request parameters.
datasets_combined	list of string	Names of the datasets joined to produce this result.
results	list of BicycleRiskResult	Each entry: { rank, district_code, district_name, total_accidents, bicycle_accidents, bicycle_accident_share_percent, accidents_per_10000_inhabitants }.

Status codes

Status	Meaning
200	Ranking returned successfully.
404	No districts matched the year and threshold.
500	Database error during bicycle risk analysis.

Example request

GET /statistics/integrated/bicycle-risk-districts/2024?limit=10&minimum_accidents=50

Example response

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/statistics/integrated/bicycle-risk-districts/2023?limit=1&minimum_accidents=100' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/statistics/integrated/bicycle-risk-districts/2023?limit=1&minimum_accidents=100
```

Server response

Code	Details
200	Response body <pre>{ "metadata": { "license_name": "Datenlizenz Deutschland - Namensnennung - Version 2.0 (DL-DE BY 2.0)", "license_code": "dl-by-de/2.0", "license_url": "http://www.dcat-ap.de/def/licenses/dl-by-de/2.0" }, "question": "Which districts had the highest bicycle accident share, and what was their official accidents per 10,000 inhabitants indicator?", "year": 2023, "minimum_accidents": 100, "datasets_combined": ["Unfallatlas accident event data", "Regional accident-rate indicator data", "AGS/GV-ISys regional reference"], "results": [{ "rank": 1, "district_code": "12054", "district_name": "Potsdam, Stadt", "total_accidents": 624, "bicycle_accidents": 365, "bicycle_accident_share_percent": 61.7, "accidents_per_10000_inhabitants": 42.5 }] }</pre>

A.6 Common Response Metadata

Every endpoint response includes the same licence metadata object, so that licence and attribution information travels with every API call rather than being looked up separately.

Field	Type	Description
license_name	string	Full name of the licence, e.g. "Datenlizenz Deutschland – Namensnennung – Version 2.0".
license_code	string	Short licence code, e.g. "dl-by-de/2.0".
license_url	string	URL of the full licence text.